ATHENS flexibility course

March 2026

Case Study YUSO:
Renewables & BESS Optimization in Europe

Agenda

- Yuso introduction
  - History and key figures
  - Market segment
  - Key projects and achievements
  - BESS Market optimization
  - C&I BESS product offering
  - YUSO and its stakeholder ecosystem

- Discussion & AOB

A short overview of our history

Since 2012 Yuso has built up expertise in all branches of the energy world, from renewables to flexibility

2012
Founded by Bart Pycke and Michel Verschuere, both managing directors to date.

2013
First energy flows.

2015
100 MWp injection portfolio achieved.

2016
1st large-scale storage with a 10 MW battery project in NL.

2017
100% green B2B energy supplier in BE.

2018
Strategic partnership with Nippon Koei for large-scale storage.

2020
Organic growth of our portfolio

2021
1st B2B flexibility service: steering of local battery storage.

2022
100 MWh battery project in BE, entry to the UK-market and the first delivery of balancing services.

2024
Extending our B2B flexibility offer with PV curtailment.

2025
Storage
E-Boilers
EV charging
PV Curt.
Wind
Heat Pumps

AGGREGATION OF RENEWABLES
LARGE-SCALE STORAGE
B2B GREEN ENERGY DELIVERY
B2B FLEXIBILITY

Yuso in figures

Turnover 2025
€ 105 million

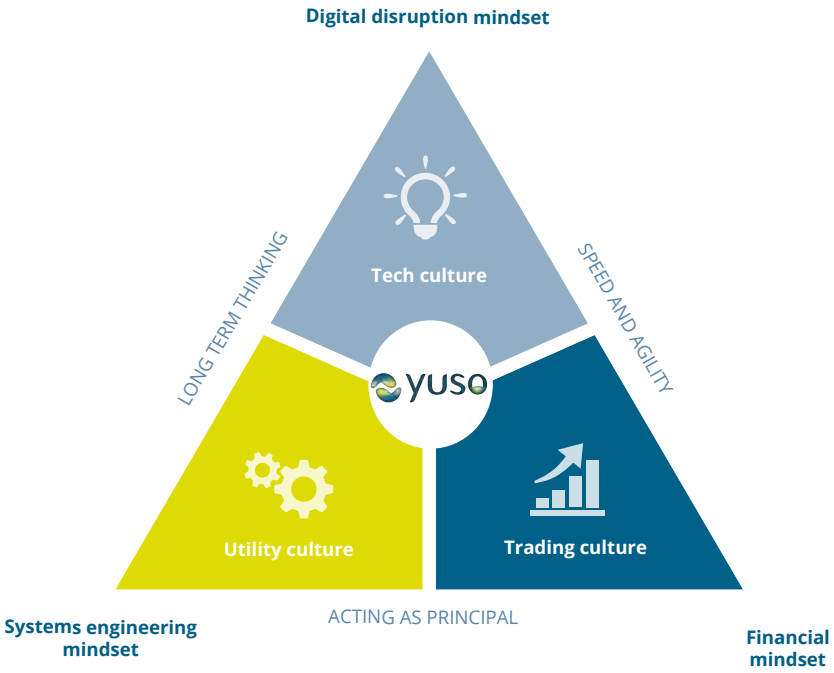
Team
55 colleagues (BE, MK)

Largest operator in BE
#1 UK BESS leaderboard

Solar PV: 450 MWp
PPA: 545 GWh
Wind: 50 MW

BESS (ALL)
BE/NL: 500 MW / 1 GWh
UK: 100 MW / 100 MWh

Countries of operation
BE, NL, UK, EE, LT, CH, FR



What we do

Yuso offers a **one-stop-shop platform** for B2B and utility-scale customers, providing **power purchase**, **supply and optimisation** services for **batteries** and **renewable power assets**.

B2B

Injection and offtake of renewable energy

A **competitive**, **flexible** and **transparent** offer for injection and offtake of renewable energy

Smart steering

High-performance assets steering, **maximizing** customers **return** while actively contributing to the system stability

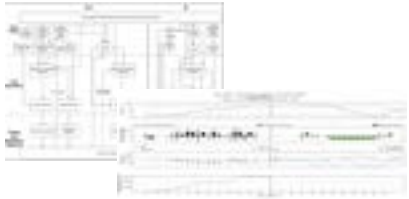
Utility Scale

Route-to-Market & PPAs

Route-to-market and **optimisation services** for utility-scale battery projects and renewable generation

Navigating through complexity

Empowering the clean energy transition through transparent, data-driven and innovative market-based solutions



Automated algorithmic strategies
Every hour, over 7m parameters are exchanged with more than 300 remote devices

Advanced forecasting models with continuous validation and retraining

Continuously changing regulatory environment and market design

yuso

Monthly invoices of BESS contracts require processing over 300'000 data points per asset

Complex business & product development processes

High reliability, low latency and cyber-secure trading platform

Projects in operation

BESS RtM under management

Ruien 25MW/ 100MWh

Balen
25MW/100MWh

AES – NL
10MW/10MWh

Deux Acren 50 MW/ 100 MWh

Tollgate – UK
50MW/50MWh

Project Delivery at Yuso (2025)

Giraffe / 10MW / 47MWh

Harmignies Energy Storage
75 MW / 300 MWh

Brecht / 25 MW / 100 MWh

103 MW / 207 MWh

113 MW / 227 MWh

220 MW / 400 MWh
undisclosed

Solid Track Record, even in highly competitive markets

Yuso is reported as the leading RtM provider for BESS based in the UK

Plants by £ / kW / Year for Jan 1 - Aug 31, 2025

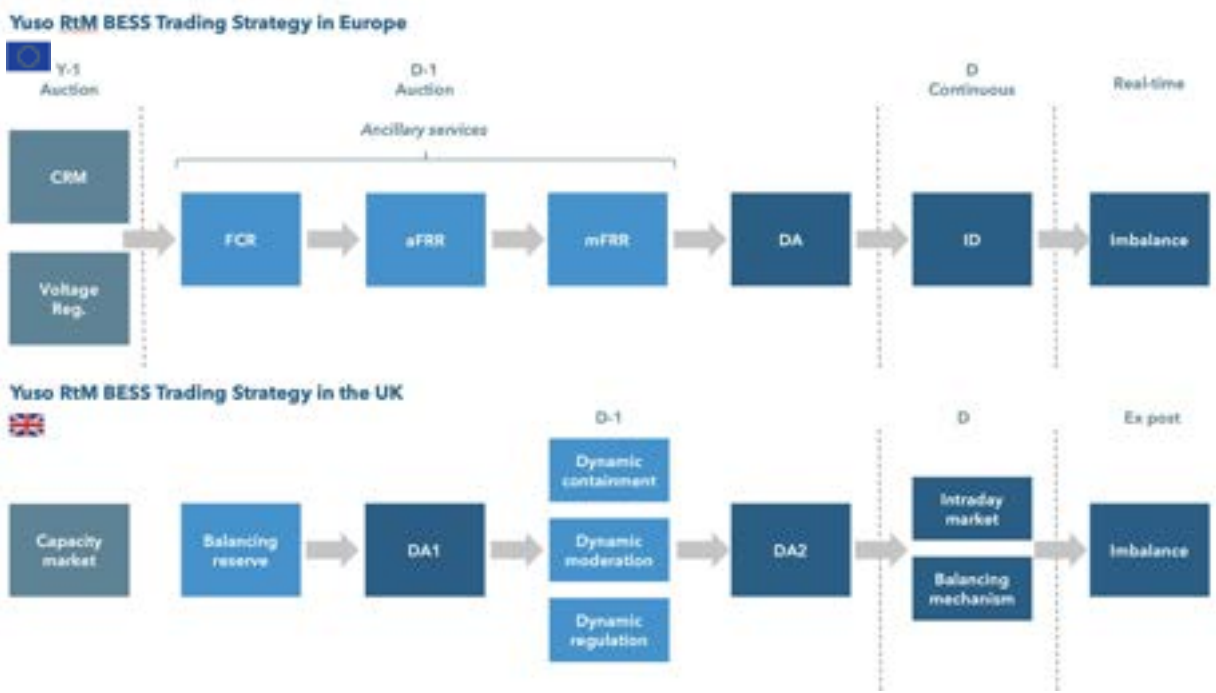
Docs | Options

Select All | Battery | Pumped Storage | Wind | Biomass | Hydro | Coal | CCGT | OCGT | Gas Recip | Supply | Other | CHP | Solar

			BESS							
ID	BM Status	Optimiser	Energy (MWh)	Duration (hrs)	No. Cycles	Net	BM	Wholesale	Wholesale Import	Wholesale Export
E_TOLLB-1	BM	yuso	56.48	1.14	211.78	71.33	2.32	10.86	-26.7	37.57
2_[illegible]	BM	Statkraft	60	1	214.27	66.49	16.36	-2.09	-28.73	26.64
2_LFLEX001	BM	Flexitricity	30	1.5	164.53	62.72	4.54	11.84	-28.38	40.22
T_RICHB-2	BM	Shell	49.9	1	363.33	67.94	7.46	35.5	-16.02	51.52
2_PFLEX001	BM	Flexitricity	49.9	1	146.8	61.37	-0.41	11.09	-17.03	28.1
T_PINFB-4	BM	EDF	27.82	1.07	211.17	60.74	5.27	12.71	-21.3	34.01
2_GSTAT005	BM	FIELD	20	1	263.91	60.44	4.86	29.34	-7.73	37.06
T_PINFB-3	BM	EDF	27.82	1.07	222.47	59.77	5.67	14.21	-20.87	35.08
T_KEMB-1	BM	EDF	49.9	1	240.47	58.96	7.92	12.29	-24.06	36.34
T_RICHB-1	BM	Shell	49.9	1	361.74	58.11	8.27	33.69	-17.77	51.46

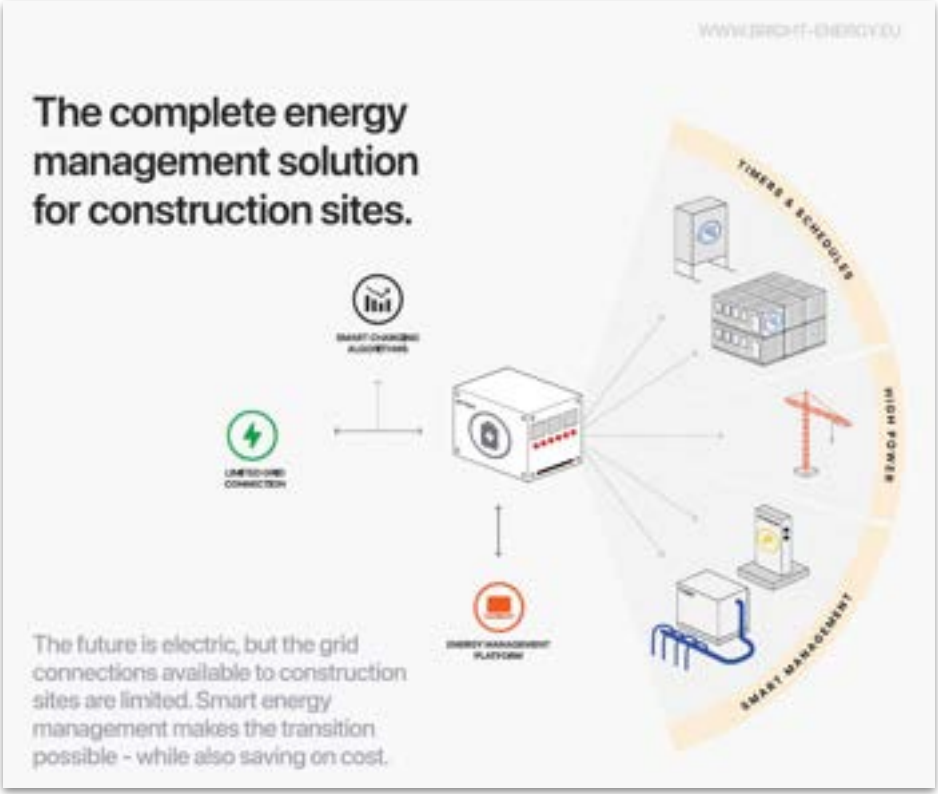
Stacking of Energy and Ancillary Markets

Yuso optimizes BESS across all available energy and ancillary markets



C&I BESS: Fast growing segment in a dynamic market

Case Study C&I BESS: Construction Yards



Case Study C&I BESS: Construction Yards (Bright-Banimmo)

200 kVA generator

→ Tower crane 1 – 135 kVA

150 kVA generator

→ Tower crane 2 – 105 kVA

44 kVA grid connection (63A)

→ Site cabins
→ Site consumption
→ EV chargers
→ Dewatering
→ …

TCO: €111,800 / year
of which €94,260 in diesel costs

Case Study C&I BESS: Construction Yards (Bright-Banimmo)

Farys – Loop Gent

Grid connection 1:

44 kVA + 87 kVA Bright Battery = 130 kVA
→ Tower crane 1
→ Tower crane 2
→ Site consumption

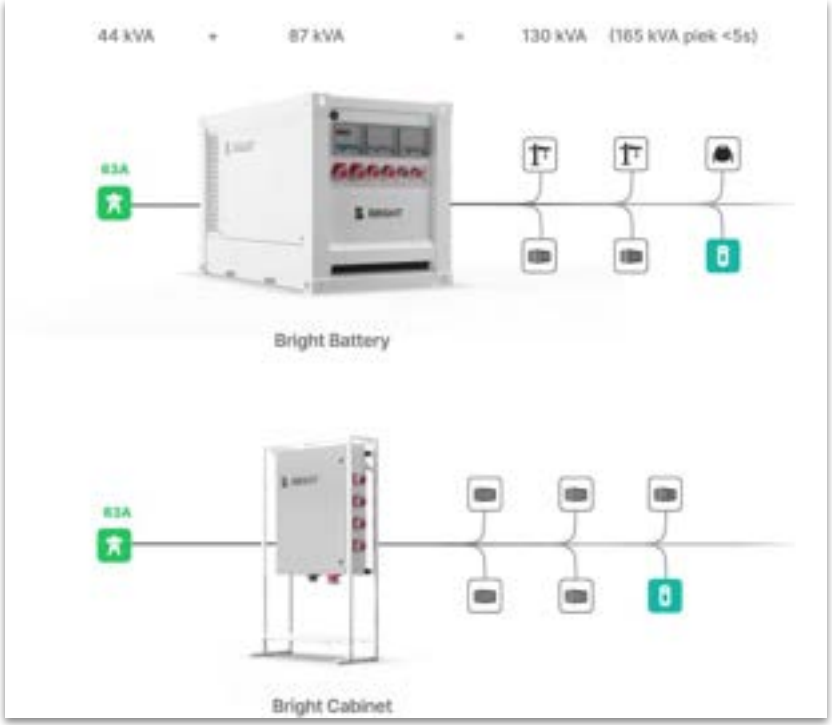
Savings vs. generator:

→ 125,000 l = €94,260
→ 405 tonnes CO_2

Grid connection 2:

44 kVA → Bright Cabinet
→ Site cabins
→ EV chargers

TCO: €39,000 / year
of which €24,900 in energy costs



BRIGHT
Electrifying construction.

Case Study C&I BESS: Construction Yards (Bright-Banimmo)

De slimme werf met Bright - case study

Farys - Loop Gent

Het **Bright Algorithme** en **Yuso** maken slimme optimalisatie mogelijk voor de laagste totale energiekost.

VAST/VARIABEL CONTRACT
€ 0,23 / kWh
energietarief

yuso
DAY AHEAD
€ 0,21 / kWh
energietarief

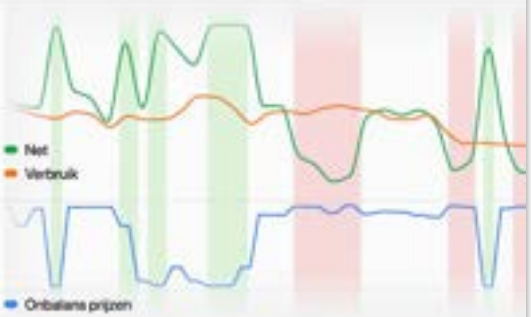
yuso
ONBALANS afnamet
€ 0,18 / kWh
energietarief

yuso
ONBALANS ingeteld
€ 0,15 / kWh
energietarief

Eigen verbruik < 0,7 kW
Besparing + € 4.700 / jaar
Door de batterij uit te schakelen buiten de uren

Timers & schedules - 24.000 kWh / jaar
Besparing + €5.548 / jaar
Door overbodig energieverbruik (zoals machines uren (dagdelen, weekenden) te elimineren

Smart charging
Besparing + € 12.768 / jaar
Door de batterij in te zetten op day-ahead en onbalansmarkt



BRIGHT
Electrifying construction.

Embedded BESS Dashboard

YUSO and its stakeholder ecosystem

Multiple stakeholders combined with a rich market environment…

ATHENS flexibility course

Discussion & AOB

We are ready
for the future...

How about you?